

Part A: Multiple choices

(26 marks)

Complete the following table by putting the correct symbol in the correct place.

Q	1	2	3	4	5	6	7	8	9	10	11	12	13
A	Aa	c	d	e	D	C	D	C	e	c	A	B	D

Q 1: Let X be a continuous random variable with pdf $f(x) = e^{-cx}$, $x \in (0, \infty)$, and zero otherwise, then $c =$

- a) 1 b) 2 c) 1.5 d) 2.5 e) 0.5

Q 2: If A and B be two events in a random experiment such that $P(A \cap \bar{B}) = 0.4$, $P(\bar{A} \cap B) = 0.1$ and $P(\bar{A} \cap \bar{B}) = 0.6$, then $P(\bar{A} \cup B) =$

- a) 0 b) 0.9 c) 0.6 d) 0.3 e) 0.8

Q 3: In the experiment of tossing a coin two times, the value of the cumulative distribution function (c.d.f.) at the interval $1 \leq x < 2$ equals

- a) 0 b) 1/4 c) 2/4 d) 3/4 e) 1

Q 4: A transmitter of some signals has 80% of its signals have no errors, 15% of its signals have one error, and 5% have two errors. Let the r.v. X be number of errors, then the expected number of errors will be

- a) 0.15 b) 0.2 c) 0.8 d) 0.35 e) 0.25

Q 5: A box in a certain supply room contains four 40-watt lightbulbs, five 60-watt lightbulbs, and six 75-watt bulbs. Suppose that 3 bulbs are randomly selected. What is the probability that exactly 2 of the selected bulbs are rated 75 watt?

- a) 0.220 b) 0.075 c) 0.148 d) 0.297 e) 0.033

Q 6: Suppose that there are five students in a classroom. What is the probability that no two students have a birthday in the same month?

- a) 0.38 b) 0.62 c) 0.22 d) 0.78 e) 0.13

Q 7: A signal is transmitted over three receivers R_1 , R_2 and R_3 whose chances of receiving it correctly are 0.5, 0.75 and 0.25, respectively. The probability that the signal will be received correctly is

- a) 0.250 b) 0.888 c) 0.125 d) 0.094 e) 0.906

Q 8: Let X be a discrete random variable with a p.d.f. $f(x) = \frac{e^{-m} 2^x}{x!}$, $x = 0, 1, 2, \dots$, then the value of m is

- a) -1 b) 1 c) 2 d) 0 e) 3

(Hint: use the Taylor series expansion of $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$)

$$e^2 \cdot e^{-m} = 1 \\ e^2 = e^m \rightarrow m = 2$$

For questions 9-10:

Let X = the number of heads appearing when a balanced coin is tossed 5 times.

Q 9: $P(X > 1 | X < 3) =$

- a) 0.23 b) 1 c) 0.385 d) 0.624 e) None

Q 10: $Var(X) =$

- a) 0.947 b) 0.053 c) 1.25 d) 0.983 e) 0.256

Q 11: A coin is flipped twice, the probability that the second flip is head given that the first flip was head is
 a) 0.25 b) 0.125 c) 0.5 d) 0.05 e) 0.75

Q 12: A random variable X takes values of 0, 1, 2, where $P(X=1) = 0.4$ and $E(X) = 1.6$, then $P(X=2)$ is
 a) 1 b) 0.6 c) 0.8 d) 0.7 e) 0.9

Q 13: Let A and B be two events such that $P(A|B) = 0.4$, $P(B|A) = 0.3$ and $P(A \cup B) = 0.8$ then $P(A \cap B) =$
 a) 0.12 b) 0.48 c) 0.58 d) 0.17 e) 0.19

Part B: Written questions

Question 1: A random variable X has a p.d.f. $f(x) = \begin{cases} (\pi/16) \cos(\pi x/8), & -4 \leq x \leq 4 \\ 0, & \text{o.w.} \end{cases}$

Find $E(X)$, $Var(X)$.

(5 marks)

$$E(X) = \int_{-4}^4 x f(x) dx = \int_{-4}^4 \frac{\pi}{16} \cdot x \cdot \cos\left(\frac{\pi x}{8}\right) dx$$

$$E(X) = \frac{\pi}{16} x \sin\left(\frac{\pi x}{8}\right) \cdot \frac{8}{\pi} + \frac{\pi}{16} \cos\left(\frac{\pi x}{8}\right) \cdot \left(\frac{8}{\pi}\right)^2$$

$$\begin{aligned} \oplus \frac{\pi}{16} x &\rightarrow \cos\left(\frac{\pi x}{8}\right) \\ \ominus \frac{\pi}{16} &\rightarrow \sin\left(\frac{\pi x}{8}\right) \cdot \frac{8}{\pi} \\ &\rightarrow -\cos\left(\frac{\pi x}{8}\right) \cdot \left(\frac{8}{\pi}\right)^2 \end{aligned}$$

$$E(X) = \frac{x}{2} \sin\left(\frac{\pi x}{8}\right) + \frac{4}{\pi} \cos\left(\frac{\pi x}{8}\right) \Big|_{-4}^4$$

$$= \left[\frac{4}{2} \sin\left(\frac{\pi}{2}\right) + \frac{4}{\pi} \cos\left(\frac{\pi}{2}\right) \right] - \left[\frac{-4}{2} \sin\left(-\frac{\pi}{2}\right) + \frac{4}{\pi} \cos\left(-\frac{\pi}{2}\right) \right]$$

$$E(X) = \text{Zero}$$

$$Var(X) = E(X^2) - (E(X))^2 = \text{Zero}$$

$$E(X^2) = \int_{-4}^4 x^2 \frac{\pi}{16} \cos\left(\frac{\pi x}{8}\right) dx$$

$$E(X^2) = Var(X) = \frac{\pi}{16} \left[x^2 \sin\left(\frac{\pi x}{8}\right) \cdot \frac{8}{\pi} + 2x \cos\left(\frac{\pi x}{8}\right) \cdot \left(\frac{8}{\pi}\right)^2 - 2 \sin\left(\frac{\pi x}{8}\right) \cdot \left(\frac{8}{\pi}\right)^3 \right]$$

$$Var(X) = 3.03 \approx 3$$

$$\begin{aligned} \oplus x^2 &\rightarrow \cos\left(\frac{\pi x}{8}\right) \\ \ominus 2x &\rightarrow \sin\left(\frac{\pi x}{8}\right) \cdot \frac{8}{\pi} \\ \oplus 2 &\rightarrow -\cos\left(\frac{\pi x}{8}\right) \cdot \left(\frac{8}{\pi}\right)^2 \\ 0 &\rightarrow -\sin\left(\frac{\pi x}{8}\right) \cdot \left(\frac{8}{\pi}\right)^3 \end{aligned}$$

Question 2: Let X be a continuous random variable with a p.d.f. $f(x) = \begin{cases} 1/2, & 0 \leq x < 1 \\ 1/4, & 1 \leq x < 3 \\ 0, & 3 \leq x < \infty \end{cases}$. Let $A = [0, 2]$, $B = [1, 4]$,

find $P(A \cup B)$ and $P(A|B)$.

(5 marks)

$$(1) P(A \cup B) = P(X \in A \cup B) = P(X \in A) + P(X \in B) - P(X \in A \cap B)$$

$$A \cup B = C$$

$$C = [$$

4
3

$$P(X \in A) = \int_{x \in A} f(x) dx = \int_0^1 \frac{1}{2} dx + \int_1^2 \frac{1}{4} dx = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

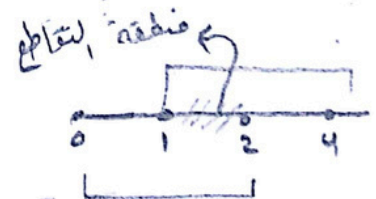
$$P(X \in B) = \int_{x \in B} f(x) dx = \int_1^3 \frac{1}{4} dx + \int_3^4 0 dx = \frac{1}{4} (3-1) = \frac{1}{2}$$

$$P(X \in A \cap B) = \int_1^2 f(x) dx = \frac{1}{4} (2-1) = \frac{1}{4}$$

$$P(A \cup B) = \frac{3}{4} + \frac{1}{2} - \frac{1}{4} = 1$$

$$(2) P(A|B) = P(X \in A|B) = \frac{P(X \in A \cap B)}{P(X \in B)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

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Q 3: Find the corresponding cdf of the pdf $f(x) = \frac{\theta^2}{\theta+1} (x+1) e^{-\theta x}$, $0 < x < \infty$.

$$F(x) = \int_{-\infty}^x f(u) du = \int_0^x f(u) du$$

$$= \int_0^x \frac{\theta^2}{\theta+1} u e^{-\theta u} du + \int_0^x \frac{\theta^2}{\theta+1} e^{-\theta u} du$$

$$\begin{aligned} & \text{① } \frac{\theta^2}{\theta+1} u \quad e^{-\theta u} \\ & \quad \downarrow \\ & \text{② } \frac{\theta^2}{\theta+1} \quad e^{-\theta u} \\ & \quad \downarrow \\ & \text{③ } 0 \quad e^{-\theta u} \end{aligned}$$

$$= \frac{\theta^2}{\theta+1} u \cdot \frac{e^{-\theta u}}{-\theta} - \frac{\theta^2}{\theta+1} \frac{e^{-\theta u}}{\theta^2} \Big|_0^x$$

$$= \frac{-\theta u e^{-\theta u}}{\theta+1} - \frac{e^{-\theta u}}{\theta+1} \Big|_0^x$$

$$\begin{aligned} & \frac{\theta^2}{\theta+1} \frac{e^{-\theta u}}{-\theta} \Big|_0^x \\ & \quad - \frac{e^{-\theta u}}{\theta+1} \Big|_0^x \\ & \quad + \frac{e^{-\theta u}}{\theta+1} \Big|_0^x \\ & \quad \frac{1}{\theta+1} (1 - e^{-\theta x}) \end{aligned}$$

5

$$\left[\frac{-\theta x e^{-\theta x}}{\theta+1} - \frac{e^{-\theta x}}{\theta+1} \right] - \left[0 - \frac{e^{-\theta \cdot 0}}{\theta+1} \right]$$

$$\frac{-1}{\theta+1} \left(+\theta x e^{-\theta x} + e^{-\theta x} - 1 \right) + \frac{1}{\theta+1} (1 - e^{-\theta x})$$

$$f(x) = -\frac{\theta x e^{-\theta x}}{\theta+1} \quad \text{X}$$



Al-Balqa' Applied University

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Mid Exam, Probability and Random Variables

Name: محمد علي الكبيسي

Lecture Time: 10:30 - 11:45

Date: 15/07/2017

Part A: Written questions

Q 1: Find the corresponding cdf of the pdf $f(x) = 2xe^{-x^2}$, $0 \leq x < \infty$.

(5 marks)

~~$$F(x) = \int_{-\infty}^x f(x) dx$$~~

$$F(x) = \int_{-\infty}^x f(u) du$$

$$F(x) = \int_{-\infty}^x 2ue^{-u^2} du$$

Let $u^2 = y \Rightarrow u=0 \rightarrow y=0$, $u=x \rightarrow y=x^2$

$$2u = \frac{dy}{du} \Rightarrow dy = \frac{dy}{2u}$$

$$\int_0^{x^2} 2u e^{-y} \frac{dy}{2u} = \int_0^{x^2} e^{-y} dy$$

$$-e^{-y} \Big|_0^{x^2} = -e^{-x^2} - (-e^0) = 1 - e^{-x^2}$$

Question 2: A random variable X has a p.d.f. $f(x) = \begin{cases} (\pi/16) \cos(\pi x/8), & -4 \leq x \leq 4 \\ 0, & \text{o.w.} \end{cases}$

(5 marks)

Find $E(X)$, $\text{Var}(X)$.

$$E(X) = \sum_{x \in R_X} x f(x)$$

$$= -4 \left(\frac{\pi}{16} \cos \frac{4\pi}{8} \right) +$$

$$E(X) = \int_{-4}^4 x f(x) dx$$

$$E(X) = \int_{-4}^4 \frac{\pi}{16} x \cos \frac{\pi x}{8} dx$$

$$E(X) = \frac{\pi}{16} \sin \left(\frac{\pi x}{8} \right) \times \frac{1}{\frac{\pi}{8}} \Big|_{-4}^4$$

$$\frac{\pi}{16} \times \frac{8}{\pi} \left(\sin \frac{4\pi}{8} - \sin \frac{-4\pi}{8} \right) = \frac{1}{2} \times 2 = 1$$

$$\text{Var } X = E X^2 - (E X)^2 = 0 + 3.031 = 3.031$$

$$E X = \int_{-4}^4 x f(x) dx = \frac{\pi}{16} \int_{-4}^4 x \cos \frac{\pi x}{8} dx$$

$$\left(\frac{8x}{\pi} \sin \left(\frac{\pi x}{8} \right) + \cos \left(\frac{\pi x}{8} \right) \times \frac{64}{\pi^2} \right) \Big|_{-4}^4$$

$$\frac{32}{\pi} + \frac{-32}{\pi} \times -1 = 0$$

$$E X^2 = \frac{\pi}{16} \int_{-4}^4 x^2 \cos \frac{\pi x}{8} dx$$

$$x^2 \cos \frac{\pi x}{8}$$

$$\downarrow$$

$$2x \sin \frac{\pi x}{8} \times \frac{8}{\pi}$$

$$\downarrow$$

$$- \cos \frac{\pi x}{8} \times \frac{8^2}{\pi^2}$$

$$\downarrow$$

$$- \sin \frac{\pi x}{8} \times \frac{8^3}{\pi^3}$$

$$E X^2 = \left(x^2 \sin \frac{\pi x}{8} \times \frac{8}{\pi} + \cos \frac{\pi x}{8} \times \left(\frac{8}{\pi} \right)^2 \right) \Big|_{-4}^4$$

$$- 2 \sin \frac{\pi x}{8} \times \left(\frac{8}{\pi} \right)^3 \Big|_{-4}^4$$

$$\frac{\pi}{16} \left[\left(16 \times \frac{8}{\pi} - 2 \left(\frac{8}{\pi} \right)^3 \right) - \left(-16 \times \frac{8}{\pi} - 2 \left(\frac{8}{\pi} \right)^3 \right) \right]$$

$$= 3.031$$

Q 3: Given the following joint probability distribution of X and Y

(6 marks)

X \ Y	1	2	3	
0	0.15	0.15	0.10	.40
1	0.10	0.10	0.20	.40
2	0.15	0.05	0.0	.20
	.40	.30	.30	1.00

a) Find $P(Y \leq X)$

b) Find $P(X \leq 1 | Y \geq 2)$

~~$(0,1)(0,2)(0,3)$~~
 ~~$\sum_R (1,1)(2,1)(2,2)$~~
 ~~$0.1 + 0.15 + 0.05 = 0.3$~~

b) $\frac{P(X \leq 1 \cap Y \geq 2)}{P(Y \geq 2)} = \frac{0.15 + 0.1 + 0.1 + 0.2}{0.15 + 0.1 + 0.05 + 0.1 + 0.2} = \frac{0.45}{0.6}$

4

= 0.75

Part B: Multiple choices

(26 marks)

Complete the following table by putting the correct symbol in the correct place.

Q	1	2	3	4	5	6	7	8	9	10	11	12	13
A	a	b	a	d	c	c	c	b	a	d	e	e	c

Q 1: Let A and B be events such that $P(A \cap \bar{B}) = 0.4$, $P(A|B) = 0.6$ and $P(A) = 0.7$, then $P(A \cup B) =$
 a) 0.9 b) 0.4 c) 0.7 d) 0.3 e) 0.5

Q 2: A random variable X takes values of 0, 1, 2, where $P(X=1) = 0.4$ and $E(X) = 1.6$, then $P(X=2)$ is
 a) 1 b) 0.6 c) 0.8 d) 0.7 e) 0.9

Q 3: Let (X, Y) be a bivariate random variable such that $E(X) = 0$, $E(Y) = 1$, $Var(X) = 1$, $Var(Y) = 4$ and $Corr(X, Y) = -1/2$, then $E(XY) =$
 a) 1 b) 4 c) -1 d) 0 e) -2

For questions 4-5, let (X, Y) be a bivariate random variable with range $R_{XY} = \{(0,1), (0,2), (1,0), (1,1), (2,0)\}$ and pdf $f(x, y) = k(x+2y)$, $(x, y) \in R_{XY}$

Q 4: The value of k so that $f(x, y)$ is a join pdf is
 a) 2/3 b) 1/3 c) 1/4 d) 1/12 e) 1/6

Q 5: $P(0 \leq X \leq 1 | Y = 1) =$
 a) 1/6 b) 5/9 c) 5/12 d) 1/4 e) 9/12

Q 6: In the experiment of tossing a coin two times, the value of the cumulative distribution function (c.d.f.) at the interval $1 \leq x < 2$ equals
 a) 0 b) 1/4 c) 2/4 d) 3/4 e) 1

Q 7: If A and B be two events in a random experiment such that $P(A \cap \bar{B}) = 0.4$, $P(\bar{A} \cap B) = 0.1$ and $P(\bar{A} \cap \bar{B}) = 0.6$, then $P(\bar{A} \cup B) =$
 a) 0 b) 0.9 c) 0.6 d) 0.3 e) 0.8

Q 8: If 5 dice are rolled, the probability that all show different faces is
 a) 0.907 b) 0.093 c) 0.001 d) 0.005 e) 0.054

Q 9: Let X be a discrete random variable with pdf $f(x) = c x$, $x = 1, 2, 3, 4$, and zero otherwise, then $c =$
 a) 0.1 b) 0.2 c) 0.05 d) 0.25 e) 2.5

Q 10: Let X be a continuous random variable with pdf $f(x) = e^{-cx}$, $x \in (0, \infty)$, and zero otherwise, then $c =$
 a) 1 b) 2 c) 1.5 d) 2.5 e) 0.5

Q 11: If X, Y be two independent discrete random variables, then $Corr(2-2X, 3+4Y) =$
 a) 0 b) 1 c) -1 d) $Cov(X, Y)$ e) $-Cov(X, Y)$

Q 12: A transmitter of some signals has 80% of its signals have no errors, 15% of its signals have one error, and 5% have two errors. Let the r.v. X be number of errors, then the expected number of errors will be
 a) 0.15 b) 0.2 c) 0.8 d) 0.35 e) 0.25

Q 13: Let X be a discrete random variable with pdf $f(x) = 0.25$, $x = 1, 2, 3, 4$, and zero otherwise, then $E(X) =$
 a) 1 b) 2 c) 1.5 d) 0.25 e) 2.5

Question 3: Let X be a discrete random variable with pdf $f(x) = \begin{cases} 0.2 & x=1,2 \\ 0.3 & x=3,4 \\ 0 & \text{otherwise} \end{cases}$

Find $\text{Var}(2-\sqrt{X})$? $\text{Var}(-\sqrt{X} + 2) = (-1)^2 \text{Var}(\sqrt{X}) = \text{Var}(\sqrt{X})$ (4 marks)

$$\text{Var}(\sqrt{X}) = E(\sqrt{X})^2 - (E\sqrt{X})^2 = E X - (E\sqrt{X})^2 \quad (1)$$

$$E X = (1)(0.2) + (2)(0.2) + (3)(0.3) + (4)(0.3) = 2.7 \quad (1)$$

$$E\sqrt{X} = \sqrt{1}(0.2) + \sqrt{2}(0.2) + \sqrt{3}(0.3) + \sqrt{4}(0.3) = 1.6 \quad (1)$$

$$\text{Var}(\sqrt{X}) = 2.7 - (1.6)^2 = 2.7 - 2.56 = 0.14 \quad (1)$$

$\Rightarrow \text{Var}(2-\sqrt{X}) = E(2-\sqrt{X})^2 - (E(2-\sqrt{X}))^2$ Using formula

Question 3: Consider the joint p.d.f. $f(x, y) = u(x)u(y)xe^{-x(y+1)}$.

a) Find the marginal p.d.f. of $f_Y(y)$.

b) Find the conditional p.d.f. $f_X(x|y)$.

$$a) f_Y(y) = \int_{-\infty}^{\infty} u(x)u(y)xe^{-x(y+1)}dx = \int_0^{\infty} u(y)xe^{-x(y+1)}dx \quad (1)$$

$$= \left[-\frac{x}{y+1}e^{-x(y+1)} - \frac{1}{(y+1)^2}e^{-x(y+1)} \right]_0^{\infty} \quad (1)$$

$$= 0 - 0 - \left[0 - \frac{1}{(y+1)^2} \right] = \frac{u(y)}{(y+1)^2} \quad (1)$$

$$b) f_X(x|y) = \frac{f(x, y)}{f_Y(y)} \quad (1)$$

$$= \frac{u(x)u(y)xe^{-x(y+1)}}{u(y)/(y+1)^2} = u(x)(y+1)^2xe^{-x(y+1)} \quad (1)$$

Question 5: Consider the function $f(x, y) = \begin{cases} b(x+y)^2, & -2 < x < 2, -3 < y < 3 \\ 0, & \text{o.w.} \end{cases}$

(5 marks)

- a) Find the constant b such that this is a joint p.d.f.
 b) Find the marginal p.d.f.'s for X and Y and then discuss the independence of the random variables.

$$(a) \int_{-2}^2 \int_{-3}^3 b(x+y)^2 dy dx = 1$$

$$\int_{-2}^2 \int_{-3}^3 b(x^2 + 2xy + y^2) dy dx = 1$$

$$b \int_{-2}^2 \left[x^2 y + \frac{2xy^2}{2} + \frac{y^3}{3} \right]_{-3}^3 dx = 1$$

$$b \int_{-2}^2 (3x^2 + 9x + 9 + 3x^2 - 9x + 9) dx = 1$$

$$b \int_{-2}^2 (6x^2 + 18) dx = 1 \Rightarrow b \left(\frac{6x^3}{3} + 18x \right) \Big|_{-2}^2 = 1$$

$$b((2)(8) + 36 - (2)(-8) + 36) = 1 \Rightarrow \underline{b = 9.6 \times 10^{-3}}$$

$$b) f_X(x) = \int_{-3}^3 b(x^2 + 2xy + y^2) dy = b(6x^2 + 18)$$

$$f_Y(y) = \int_{-2}^2 b(x^2 + 2xy + y^2) dx$$

$$= b \left(\frac{x^3}{3} + x^2 y + y^2 x \right) \Big|_{-2}^2$$

$$= b \left(\frac{8}{3} + \frac{8}{3} + 4y - 4y + 2y^2 + 2y^2 \right)$$

$$= b \left(\frac{16}{3} + 4y^2 \right)$$

$$f_X(x) \cdot f_Y(y) \neq f(x, y)$$

$\therefore$ not indep.

Part A: Written questions

Q 1: Find the corresponding cdf of the pdf $f(x) = 2xe^{-x^2}$, $0 \leq x < \infty$

(4 marks)

$$\begin{aligned}
 F(x) &= \int_0^x 2u e^{-u^2} du \quad (1) \\
 &= \int_0^{x^2} 2\alpha e^{-\alpha} \frac{d\alpha}{2\alpha} = -e^{-\alpha} \Big|_0^{x^2} \quad (1) \\
 &= -(e^{-x^2} - 1) \\
 &= 1 - e^{-x^2} \quad (1)
 \end{aligned}$$

$y = u^2 \Rightarrow dy = 2u du$
 $du = \frac{dy}{2u}$
 $u = x \Rightarrow y = x^2$

Question 2: A random variable X has a p.d.f. $f(x) = \begin{cases} (\pi/16) \cos(\pi x/8), & -4 \leq x \leq 4 \\ 0, & \text{o.w.} \end{cases}$

Find $E(X)$, $Var(X)$.

(4 marks)

$$E(X) = \int_{-4}^4 \frac{\pi}{16} x \cos\left(\frac{\pi}{8}x\right) dx$$

$$= \left[\frac{8}{\pi} x \sin\left(\frac{\pi}{8}x\right) + \frac{8^2}{\pi^2} \cos\left(\frac{\pi}{8}x\right) \right]_{-4}^4 \quad (2)$$

$$= \frac{32}{\pi} \sin\left(\frac{\pi}{2}\right) + \frac{8^2}{\pi^2} \cos\left(\frac{\pi}{2}\right) + \frac{32}{\pi} \sin\left(-\frac{\pi}{2}\right) - \frac{8^2}{\pi^2} \cos\left(-\frac{\pi}{2}\right)$$

$$= 0 \cdot 4$$

$$E(X^2) = \int_{-4}^4 \frac{\pi}{16} x^2 \cos\left(\frac{\pi}{8}x\right) dx$$

$$= \frac{\pi}{16} \left[\frac{8}{\pi} x^2 \sin\left(\frac{\pi}{8}x\right) + 2 \frac{8^2}{\pi} x \cos\left(\frac{\pi}{8}x\right) - 2 \frac{8^3}{\pi^2} \sin\left(\frac{\pi}{8}x\right) \right]_{-4}^4 \quad (1)$$

$$= \frac{\pi}{16} \left[\frac{8}{\pi} (4)^2 \sin\left(\frac{\pi}{2}\right) + 2 \frac{8^2}{\pi} (4) \cos\left(\frac{\pi}{2}\right) - 2 \frac{8^3}{\pi^2} \sin\left(\frac{\pi}{2}\right) - \frac{8}{\pi} (4)^2 \sin\left(-\frac{\pi}{2}\right) - 2 \frac{8^2}{\pi} (4) \cos\left(-\frac{\pi}{2}\right) + 2 \frac{8^3}{\pi^2} \sin\left(-\frac{\pi}{2}\right) \right]$$

$$= \frac{\pi}{16} \left[32 \frac{8}{\pi} - 4 \frac{8^3}{\pi^2} \right] = 16 - \frac{128}{\pi^2}$$

$$\begin{aligned}
 Var(X) &= E(X^2) - (E(X))^2 \\
 &= 16 - \frac{128}{\pi^2} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 &\cos\left(\frac{\pi}{8}x\right) \\
 &\frac{8}{\pi} \sin\left(\frac{\pi}{8}x\right) \\
 &-\frac{8^2}{\pi^2} \cos\left(\frac{\pi}{8}x\right)
 \end{aligned}$$

$$\begin{aligned}
 &x^2 \cos\left(\frac{\pi}{8}x\right) \\
 &2x \cdot \frac{8}{\pi} \sin\left(\frac{\pi}{8}x\right) \\
 &2 \cdot \frac{8^2}{\pi^2} \cos\left(\frac{\pi}{8}x\right) \\
 &0 \cdot \frac{8^3}{\pi^3} \sin\left(\frac{\pi}{8}x\right)
 \end{aligned}$$

Question 3: Let X be a discrete random variable with pdf $f(x) = \begin{cases} 0.2 & , x=1,2 \\ 0.3 & , x=3,4 \\ 0 & , \text{otherwise} \end{cases}$

Find $Var(2-\sqrt{X})$? $Var(-\sqrt{X} + 2) = (-1)^2 Var(\sqrt{X}) = Var(\sqrt{X})$ (4 marks)

$$Var(\sqrt{X}) = E(\sqrt{X})^2 - (E\sqrt{X})^2 = E X - (E\sqrt{X})^2 \quad (1)$$

$$E X = (1)(0.2) + (2)(0.2) + (3)(0.3) + (4)(0.3) = 2.7 \quad (1)$$

$$E\sqrt{X} = \sqrt{1}(0.2) + \sqrt{2}(0.2) + \sqrt{3}(0.3) + \sqrt{4}(0.3) = 1.6 \quad (1)$$

$$Var(\sqrt{X}) = 2.7 - (1.6)^2 = 2.7 - 2.56 = 0.14 \quad (1)$$

$\Rightarrow Var(2-\sqrt{X}) = E(2-\sqrt{X})^2 - (E(2-\sqrt{X}))^2$
 Using formula

Question 3: Consider the joint p.d.f. $f(x, y) = u(x)u(y)xe^{-x(y+1)}$.

a) Find the marginal p.d.f. of $f_y(y)$.

b) Find the conditional p.d.f. $f_x(x|y)$.

$$a) f_y(y) = \int_{-\infty}^{\infty} u(x)u(y)xe^{-x(y+1)}dx = \int_0^{\infty} u(y)xe^{-x(y+1)}dx \quad (1)$$

$$= \left[\frac{-x}{y+1} e^{-x(y+1)} - \frac{1}{(y+1)^2} e^{-x(y+1)} \right]_0^{\infty} \quad (1)$$

$$= 0 - 0 - \left[0 - \frac{1}{(y+1)^2} \right] = \frac{1}{(y+1)^2} \quad (1)$$

$$b) f_x(x|y) = \frac{f(x, y)}{f_y(y)} \quad (1)$$

$$= \frac{u(x)u(y)xe^{-x(y+1)}}{u(y)/(y+1)^2} = u(x)(y+1)^2xe^{-x(y+1)} \quad (1)$$

SUBJECT

احصاء

SECTION

MIDTERM EXAM-2017

فصل الثاني

Q1 find The c.d.f , for The Rayleigh density

$$\text{that is given by } \Rightarrow f(x) = \begin{cases} 2\lambda x e^{-\lambda x^2}, & 0 \leq x < \infty \\ 0, & \end{cases}$$

SOL:-

$$f(x) = \int_0^x 2\lambda u e^{-\lambda u^2} du \quad (1)$$

$$= \int_0^{-\lambda x^2} 2\lambda u e^y \frac{dy}{-2\lambda u} \quad (1)$$

$$= -(e^y]_0^{-\lambda x^2}) \quad (1)$$

$$= -(e^{-\lambda x^2} - 1)$$

$$= 1 - e^{-\lambda x^2} \quad (1)$$

5 marks

* لا تنسى التعويض

$$y = -\lambda u^2$$

$$dy = -2\lambda u du \quad (1)$$

$$u=0 \Rightarrow y=0$$

$$u=x \Rightarrow y = -\lambda x^2$$

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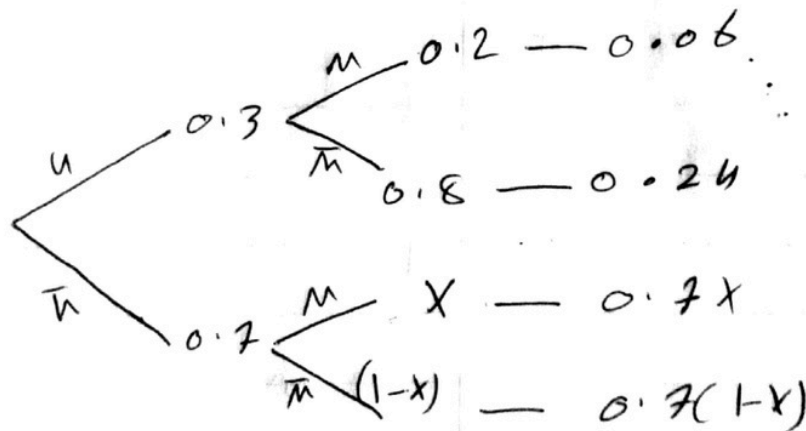
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[Q2] A transmitter Produced 30% of radioactive signals with source noise and 70% of good signals that may effect by external noise 20% of the source bad signals can be managed using a good receiver . if the Proportion of signals that be managed out of all signal transmitted away from The transmitter is 60% , find the Proportion of signals that could be managed using the reciver such that The ~~original signal~~ original signals were good when transmitted from The Transmitter ?

Smarts

SOL :-



$$0.6 = P(M) = 0.06 + 0.7X \quad (2)$$

$$X = P(M/\bar{n}) = \frac{0.54}{0.70} = 0.77 \quad (1)$$

SUBJECT**SECTION** 2014 - CIV JPE $\rightarrow$

Q 3 Each of two transmitters worked for 100 h hours, with different wave length. Three receivers are supposed to receive their signals based on their wave length. The following table shows the connection times (in hours) between the transmitters and receivers.

X		Transmitters		Total
		T ₁	T ₂	
1	Receiver 1	15	50	65
2	Receiver 2	60	30	90
3	Receiver 3	25	20	45
Total		100	100	200

6 marks

find?

- The proportion of times for each receiver such that the signals ~~where~~ were produced from the transmitters T₁ and T₂?
- find the conditional P.d.f.s $f_X(X|T_1)$ and $f_X(X|T_2)$?
- find the P.d.f for $f_X(X)$?

SUBJECT**SECTION**

2017-18 L'Ap'p' n°

Sol:-
of Q3

$$\begin{array}{l|l} \text{a)} & P(X=1|T_1) = 0.15 \quad P(X=1|T_2) = 0.5 \\ & P(X=2|T_1) = 0.6 \quad P(X=2|T_2) = 0.3 \\ & P(X=3|T_1) = 0.25 \quad P(X=3|T_2) = 0.2 \end{array}$$

(2)

$$\text{b)} \quad f(X|T_1) = 0.15 \delta(X-1) + 0.6 \delta(X-2) + 0.25 \delta(X-3) \quad (1)$$

$$f(X|T_2) = 0.5 \delta(X-1) + 0.3 \delta(X-2) + 0.2 \delta(X-3) \quad (1)$$

$$\begin{aligned} \text{c)} \quad P(X=1) &= P(X=1|T_1)P(T_1) + P(X=1|T_2)P(T_2) \\ &= 0.15 \times 0.5 + 0.5 \times 0.5 \\ &= 0.325 \quad (0.5) \end{aligned}$$

$$\begin{aligned} P(X=2) &= P(X=2|T_1)P(T_1) + P(X=2|T_2)P(T_2) \\ &= (0.6)(0.5) + (0.3)(0.5) = 0.45 \quad (0.5) \end{aligned}$$

$$\begin{aligned} P(X=3) &= (0.25)(0.5) + (0.2)(0.5) = \cancel{0.225} \\ &= 0.225 \quad (0.5) \end{aligned}$$

$$f_X(X) = 0.325 \delta(X-1) + 0.45 \delta(X-2) + 0.225 \delta(X-3) \quad (0.5)$$

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SUBJECT

SECTION

2017 - 12 Dec 20

Q4 Let The r.v. X follows The exponential distribution with p.d.f

$$f_X = \begin{cases} \lambda e^{-\lambda x}, & 0 < x < \infty \\ 0, & \text{o.w} \end{cases}$$

find :- a) λ

b) $E(X)$, $\text{Var}(X)$?

5 marks

Solu:-

$$a) \lambda \int_0^{\infty} e^{-\lambda x} dx = 1 \quad \text{---} \rightarrow \lambda \text{ is arbitrary} \quad (1)$$

$$b) E(X) = \lambda \int_0^{\infty} x e^{-\lambda x} dx$$

$$= \lambda \left(\frac{-x}{\lambda} e^{-\lambda x} - \frac{1}{\lambda^2} e^{-\lambda x} \right) \Big|_0^{\infty} = \frac{1}{\lambda} \quad (2)$$

$$E(X^2) = \lambda \int_0^{\infty} x^2 e^{-\lambda x} dx = \lambda \left(\frac{-x^2}{\lambda} e^{-\lambda x} - \frac{2}{\lambda} x e^{-\lambda x} - \frac{2}{\lambda^2} e^{-\lambda x} \right) \Big|_0^{\infty}$$

$$= \lambda \left(\frac{2}{\lambda^2} \right) = \frac{2}{\lambda} \quad (1)$$

$$\text{Var } X = E(X^2) - (E(X))^2 = \frac{2}{\lambda} - \left(\frac{1}{\lambda} \right)^2 = \frac{2\lambda - 1}{\lambda^2} \quad (1)$$

SUBJECT

SECTION 2014 - 21st Dec 20

Q5 Consider The joint P.d.f $f(x,y) = \begin{cases} \frac{5}{16} x^2 y, & 0 < y < x < 2 \\ 0 & \text{otherwise} \end{cases}$

find the marginal P.d.f's for x, y and then

discuss the independence of the random v's?

sol:-

$$f_x(x) = \int_0^2 \frac{5}{16} x y^2 dy = \left[\frac{5}{16} x^2 \frac{y^3}{3} \right]_0^2$$
$$= \frac{4(5)}{16(3)} x^2$$

$$f_y(y) = \int_0^2 \frac{5}{16} x^2 y dx = \left[\frac{5}{16} y \frac{x^3}{3} \right]_0^2$$
$$= \frac{5(8)}{(16)(3)} y$$

$$f(x,y) \neq f_x(x) f_y(y) \text{ since}$$

$$\frac{5}{16} x^2 y \neq \frac{4(5)}{(16)(3)} x^2 \cdot \frac{5(8)}{16(3)} y = \frac{5}{3} \frac{5}{16} x^2 y$$

so, not indep.

SUBJECT

SECTION

2017 - 2018

Multiple choices

Q1:- Let A, B two event such that $P(A) = 4/7$
 $P(A \cap B) = 3/7$ and $P(A|B) = 2/3$, The $P(A|\bar{B})$??

a) 0.1

b) 0.4

c) 0.3

e) 0.5

Q2:- a telecommunication company has noticed that 80% of their signals have no error when they reach some end point. 15% of their signals have one error, and 5% have two errors, if you buy a new receiver from this company, the expected number of errors in this service will be?

a) 0.15

b) 0.2

c) 0.8

d) 0.35

e) 0.25

Q3) Let (X, Y) be a bivariate random variable such that $\text{Var}(X) = 9$, $\text{Var}(Y) = 4$ and $\text{Cov}(X, Y) = 3$. Then $\text{Corr}(1 - 3X, 3Y - 1) = ?$

a) $-1/2$

b) $-1/4$

c) $1/2$

d) $1/4$

e) 1

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SUBJECT**SECTION**

2017 - 12 Dec no

Q4:- Let X be a discrete r.v with
P.d.f $f(x) = 0.5$, $x = 1, 2, 3, 4$, and zero other
then $E(X) = ?$

a) 1

b) 2

c) 1.5

d) 0.25

e) 2.5

for Q 5, 6

$x \backslash y$	0	1	2	3	
0	$1/8$	$1/8$	0	0	
1	0	$2/8$	$2/8$	0	
2	0	0	$1/8$	$1/8$	

Q5:- The Probability that X is greater
Than y ($P(X > y)$) = ?

a) 0

b) $1/8$ c) $2/8$ d) $3/8$ e) $4/8$

Q6) $P(0 \leq X \leq 1 | y=1) =$

a) 0

b) $2/8$ c) $4/8$ d) $6/8$

e) 1

SUBJECT

SECTION

2014 - 15th Dec 20

Q7:- in the experiment of Tossing a coin Two times, The value of The cumulative distribution function (c.d.f) at the interval $1 \leq X < 2 = ?$

a) 0

b) $1/4$

c) $2/4$

d) $3/4$

e) 1

Q8) Let A, B be events such that

$P(A) = 0.45$, $P(B) = 0.5$ and $P(\bar{A} \cap \bar{B}) = 0.2$
compute $P(\bar{A} \cap \bar{B})$

a) 0.8

b) 0.15

c) 0.95

d) 0.23

e) none

Q1:- Find The corresponding cdf of The
Pdf $f(x) = \frac{1}{\theta} e^{-x/\theta}$, $0 \leq x < \infty$?

Solu:-

$$F(x) = \int_0^x \frac{1}{\theta} e^{-y/\theta} dy$$

$$= \frac{1}{\theta} \frac{e^{-y/\theta}}{-1/\theta}$$

$$= -e^{-y/\theta} \Big|_0^x$$

$$= -e^{-x/\theta} - (-e^0)$$

$$= -e^{-x/\theta} + 1$$

$$F(x) = 1 - e^{-x/\theta}$$

Q2:- for the Rayleigh density that is given

$$\text{by } f(x) = \begin{cases} 2\lambda x e^{-\lambda x^2}, & 0 \leq x < \infty \\ 0, & \text{o.w.} \end{cases}$$

find $E(X)$, $\text{Var}(X)$?

sol:-

$$E_x = \int_0^{\infty} x f(x) \cdot dx = \int_0^{\infty} x^2 2\lambda e^{-\lambda x^2} \cdot dx$$

$$= 2\lambda \left(-2\lambda x^2 e^{-\lambda x^2} - 8\lambda^2 x e^{-\lambda x^2} - 16\lambda^3 e^{-\lambda x^2} \right) \Big|_0^{\infty}$$

$$= -(16\lambda^3 e^{-\infty} - 16\lambda^3 e^0) 2\lambda$$

$$E_x = 16\lambda^3 (2\lambda)$$

$$E_x = 32\lambda^4$$

x^2	$e^{-\lambda x^2}$
$2x$	$-2\lambda e^{-\lambda x^2}$
2	$-4\lambda^2 x e^{-\lambda x^2}$
0	$-8\lambda^3 e^{-\lambda x^2}$

$$\text{Var } X = E_x^2 - (E_x)^2$$

end

$$E X^2 = \int_0^{\infty} x^3 2 \lambda e^{-\lambda x^2} dx$$

$$= 2\lambda \left(-2\lambda x^3 e^{-\lambda x^2} - 12\lambda^2 x^2 e^{-\lambda x^2} - 48\lambda^3 x e^{-\lambda x^2} - 96\lambda^4 e^{-\lambda x^2} \right) \Big|_0^{\infty}$$

$$= 2\lambda (0 - (-96\lambda^4))$$

$$= 192 \lambda^5 = E X^2$$

$$\text{Var } X = E X^2 - (E X)^2$$

$$\text{Var } X = 192 \lambda^5 - 32 \lambda^4$$

$$\begin{array}{l} x^3 \cdot e^{-\lambda x^2} \\ \ominus \quad 3x^2 \cdot -2\lambda e^{-\lambda x^2} \\ \ominus \quad 6x \cdot 4\lambda^2 e^{-\lambda x^2} \\ \oplus \quad 6 \cdot -8\lambda^3 e^{-\lambda x^2} \\ \ominus \quad 0 \cdot 16\lambda^4 e^{-\lambda x^2} \end{array}$$

Q3:- Let X be a discrete random variable with Pdf

$f(x)$	$\left\{ \begin{array}{ll} 0.2 & , x = 1, 2 \\ 0.3 & , x = 3, 4 \\ 0 & , 0.4 \end{array} \right.$	3
--------	---	---

find $\text{var}(2 - \sqrt{x})$?

Solu:-

$$\text{Var}(2 - \sqrt{x}) = \text{Var}(-\sqrt{x} + 2)$$

$$= (-1)^2 \text{Var}(\sqrt{x})$$

$$= \text{Var} \sqrt{x}$$

$$= E(\sqrt{x})^2 - (E\sqrt{x})^2$$

$$= E - (E\sqrt{x})^2$$

$$= \sum x f(x) - \sum \sqrt{x} f(x)$$

$$= (1)(0.2) + (2)(0.2) + (3)(0.3) + (4)(0.3)$$

$$= -\sqrt{1}(0.2) + \sqrt{2}(0.2) - \sqrt{3}(0.3) - \sqrt{3}(0.3)$$

$$\boxed{\text{Var}(2 - \sqrt{x}) = E(2 - \sqrt{x})^2 - (E(2 - \sqrt{x}))^2}$$

Qu:- consider the joint P.d.f $f(x,y)$

$$f(x,y) = \begin{cases} \frac{5}{16} x^2 y, & 0 < y < x < 2 \\ 0, & \text{o.w} \end{cases}$$

find The marginal P.d.f's for x and y then discuss The independence of The random variables?

Solu:- $f_x(x) = \int_{R_y} f(x,y) dy$

$$= \int_0^2 \frac{5}{16} x^2 y \cdot dy = \frac{5}{16} x^2 \frac{y^2}{2} \Big|_0^2$$

$$\frac{5}{16} x^2 (2-0) = \frac{5}{16} x^2 \times 2 = \boxed{\frac{5}{8} x^2 = f_x(x)}$$

$$f_y(y) = \int_0^2 \frac{5}{16} x^2 y \cdot dx = \frac{5}{16} \frac{x^3}{3} y \Big|_0^2$$

$$= \frac{5}{16} y \left(\frac{8}{3} - 0 \right) = \frac{5}{16} \times \frac{8}{3} y = \frac{5}{6} y$$

$$\boxed{f_y(y) = \frac{5}{6} y}$$

$f_y(y) \times f_x(x) \neq f(x,y)$, Then is ~~not~~ independent

Multiple choices :-

- 1) Let A, B be events that $P(A \cap B) = 0.4$
 $P(A|B) = 0.6$ and $P(A) = 0.7$ then $P(A \cup B) =$
a) 0.9 b) 0.4 c) 0.7 d) 0.3 e) 0.5
- 2) A quality controller of a telecommunication company has noticed 70% of their signals have no defective, 20% of their signals have one defective, and 10% have two defective, If you test the signals transmitted from this company, then the expected number of defective will be
a) 0.45 b) 0.4 c) 0.5 d) 0.35 e) 0.25
- 3) Let (X, Y) random variable such that $E(X) = 2.5$
 $E(Y) = 4$, $\text{Var}(X) = 4$, $\text{Var}(Y) = 1$ and $E(XY) = 12$
Then $\text{Corr}(2X-1, 1-Y) =$
a) 0.5 b) 0 c) 1 d) -1 e) -0.5

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NOTE

$x \backslash y$	1	2	3	4
-1	0.01	0.09	0.1	0.2
0	0.05	0.15	0.01	0.01
1	0.09	0.1	0.11	0.08

4) Then $E(XY) =$

a) 0.35

b) -0.51

c) -0.35

d) 0.5

e) 0.29

5) $P(0 \leq X \leq 1 | Y=1) =$

a) 0.09

b) 0.05

c) 0.14

d) 0.93

e) 0.15

6) In the experiment of tossing a coin two times
The value of the cumulative distribution function
(C.D.F) at the interval $1 \leq X < 2$ equals

a) 0

b) $1/4$ c) $2/4$ d) $3/4$

e) 1

7) if A and B be two events in a random exper-
such that $P(A \cap B) = 0.4$, $P(\bar{A} \cap B) = 0.1$ and $P(\bar{A} \cap \bar{B}) = 0.6$
Then $P(\bar{A} \cup B) =$

a) 0

b) 0.9

c) 0.6

d) 0.3

e) 0.8

Don't miss the change..

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8) if 5 dice are rolled, the probability that all show different faces is :-

- a) 0.907 , **b) 0.093** c) 0.001 d) 0.005
e) 0.054

9) Let X be a discrete random var. with pdf $f(x) = c x$, $x = 1, 2, 3, 4$ and zero, find c ?

- a) 0.1** b) 0.2 c) 0.05 d) 0.25 e) 2.5

10) Let X be a continuous random variable with pdf $f(x) = e^{-cx}$, $x \in (0, \infty)$ and zero find c ?

- a) 1** b) 2 c) 1.5 d) 2.5 e) 0.5



(3h)

Lecture Time: 11-2.12

Date: 17/03/2019

(18 marks)

Part A: Written questions

Question 1: Let X be a discrete random variable with pdf $f(x) = \begin{cases} 0.2 & , x=1,2 \\ 0.3 & , x=3,4 \\ 0 & , \text{otherwise.} \end{cases}$

Find $\text{Var}(2 - \sqrt{X})$?

(5 marks)

$$\text{Var} X = E X^2 - (E X)^2$$

$$\begin{aligned} E X &= 0 p(0) + 1 p(1) + 2 p(2) + 3 p(3) + 4 p(4) \\ &= 0 + 0.2 + 2 \times 0.2 + 3 \times 0.3 + 4 \times 0.3 \\ &= 2.7 \end{aligned}$$

$$\begin{aligned} E X^2 &= 0 p(0) + 1 p(1) + 4 p(2) + 9 p(3) + 16 p(4) \\ &= 0 + 0.2 + 0.8 + 2.7 + 4.8 \\ &= 8.5 \end{aligned}$$

$$\text{Var} X = 8.5 - (2.7)^2 = 1.21$$

$$\begin{aligned} \text{Var}(2 - \sqrt{X}) &= (-1) \times \sqrt{\text{Var} X} \\ &= -1.1 \end{aligned}$$